

Challenge #3

Step 1 - Access the dataset

I chose SQL to analyse the data so I accessed on Google BigQuery.

Step 2 - Understand the data collection goal

The goal is to retrieve a dataset that can be used to identify countries with an old population of 65 years and above, which is the company's target market. It has been mentioned in the scenario that two things that should be found from the dataset are:

1. Identify the five most promising major markets
2. Identify five developing countries to which the company could potentially supply these devices.

From this goal, we must find some data, namely the population and income of each country.

Step 3 - Collect the relevant data

The relevant data to answer the objectives are:

3. bigquery-public-data.world_bank_global_population.population_by_country
4. bigquery-public-data.world_bank_wdi.country_summary.

Step 4 - Collate your report

I took 5 features/variables: 3 from the country summary dataset and 2 from the population dataset. The query retrieves the top 100 countries, along with their population for 2018, special notes, and income groups. This query ensures that only countries with valid (non-zero) income groups are included, and the results are sorted by most population.

```
SELECT
  pop.country,
  country.country_code,
  country.special_notes,
  country.income_group,
  pop.year_2018
FROM `bigquery-public-data.world_bank_wdi.country_summary` AS country
LEFT JOIN `bigquery-public-data.world_bank_global_population.population_by_country`
AS pop
ON country.country_code = pop.country_code
WHERE country.income_group IS NOT NULL
ORDER BY pop.year_2018 DESC
LIMIT 100;
```

If we want to do a more specific search for 5 developing countries, we can change the query by specifying the search income = High income because the income of developing countries is identical to that of their population, which is in the high category.

```
SELECT
  pop.country,
  country.country_code,
  country.special_notes,
  country.income_group,
  pop.year_2018
FROM `bigquery-public-data.world_bank_wdi.country_summary` AS country
LEFT JOIN `bigquery-public-data.world_bank_global_population.population_by_country` AS pop
ON country.country_code = pop.country_code
WHERE country.income_group IN ('High income')
ORDER BY pop.year_2018 DESC
LIMIT 5;
```

The five developing countries with the most promising large markets are the United States, Japan, Germany, France, and the United Kingdom.

